SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES

DEPARTMENT OF FINANCIAL ENGINEERING

Modelling Physical Climate Risk in IFRS 9 ECL Frameworks: A Two-Regime Markov-Switching Approach to Credit Migration in Zimbabwe's Microfinance Sector

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THIS RESEARCH IS SUBMITTED TO THE DEPARTMENT OF FINANCIAL ENGINEERING, SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES, IN PARTIAL FULFILLMENT OF THE BACHELOR OF TECHNOLOGY HONOURS DEGREE IN FINANCIAL ENGINEERING

2026

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CHAPTER 1

Introduction

1.1 Background to the Study

1.1.1 The Global Imperative for Dynamic Credit Risk Modeling

The foundation of modern financial risk management is the requirement for financial institutions to maintain adequate capital buffers against anticipated losses. This requirement was fundamentally transformed by the introduction of International Financial Reporting Standard 9 (IFRS 9), which replaced the incurred loss model (IAS 39) with a forward-looking Expected Credit Loss (ECL) framework in 2018 (Novotny-Farkas, 2016). The IFRS 9 ECL model mandates that institutions must incorporate "reasonable and supportable information" regarding forecasts of future economic conditions into the calculation of the Probability of Default (PD) and Lifetime ECL (IASB, 2014; Cohen and Edwards, 2017). This move has shifted credit risk modeling from a static, "Through-the-Cycle" (TTC) approach to a dynamic, "Point-in-Time" (PIT) methodology (European Banking Authority, 2023). Crucially, the latest regulatory reviews emphasize that institutions must now accelerate efforts to integrate climate and sustainability-related risks into their ECL outputs, recognizing climate change as a systemic, financial risk driver (European Banking Authority, 2023). This global mandate establishes the necessity for dynamic models that can link external, non-financial factors to credit loss.

1.1.2 Climate Risk and Microfinance Vulnerability in Zimbabwe

In Sub-Saharan Africa, and specifically in Zimbabwe, the economy is highly sensitive to weather-related shocks, primarily drought, which acts as a powerful, non-linear driver of systemic financial risk (Klomp, 2014; Castellani and Cincinelli, 2015). Microfinance Institutions (MFIs) are uniquely vulnerable because their client base—comprised largely of small-scale farmers and informal sector entrepreneurs—has a direct, inelastic link between hydrological performance and cash flow (Chikalipah, 2018). Drought shocks lead to widespread crop failure and reduced non-farm activity, resulting in highly covariant (correlated) loan defaults across the entire portfolio (Castellani and Cincinelli, 2015). These losses destroy the MFIs' capital reserves, reducing the subsequent supply of credit and exacerbating financial exclusion (Collier and Dercon, 2014). Consequently, managing this physical climate risk is the single most critical challenge to the long-term viability and financial stability of the Zimbabwean microfinance sector, a fact now recognized by the Reserve Bank of Zimbabwe's renewed focus on risk management guidelines (Reserve Bank of Zimbabwe, 2024).

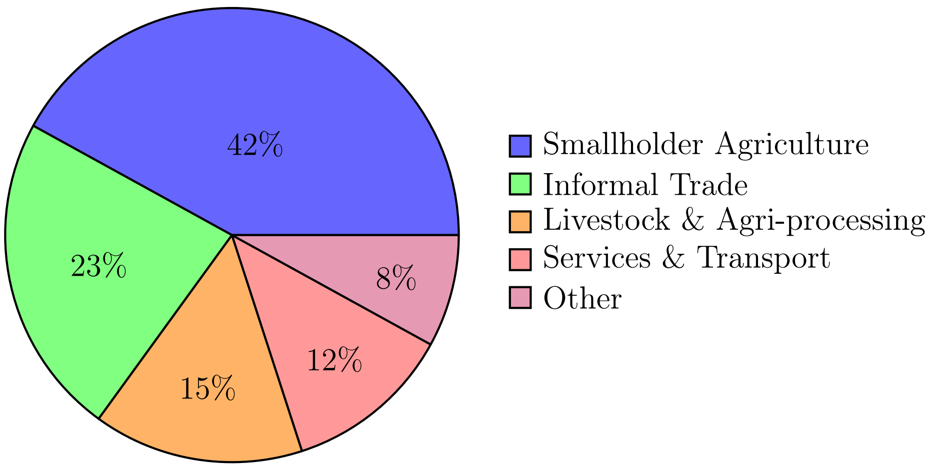

Figure 1: Estimated Composition of Zimbabwean MFI Client Base by Sector (Reserve Bank of Zimbabwe, 2024)

1.1.3 Limitations of Current Credit Risk Models

Despite the established link between climate cycles and Non-Performing Loans (NPLs), local financial institutions, particularly MFIs, rely predominantly on static or basic econometric models for credit risk assessment (Chikalipah, 2018). Studies on Zimbabwean banks have confirmed that static models are insufficient, noting that credit risk is largely explained by the volatile macroeconomic environment rather than microeconomic variables (Nyamutowa and Masunda, 2013). Furthermore, while the Reserve Bank of Zimbabwe (RBZ) requires sound risk management, the MFI sector often lacks the sophisticated credit infrastructure necessary to implement complex internal ratings-based models (Reserve Bank of Zimbabwe, 2024). The fundamental limitation is that existing local models are time-homogeneous (TTC); they estimate an average probability of default that does not account for the probability of the economy (or the agricultural sector) switching abruptly into a severe drought state, as required by the forward-looking demands of IFRS 9.

1.1.4 The Methodological Gap and Proposed Solution

The confluence of the global IFRS 9 mandate and the local vulnerability of MFIs creates a clear, urgent methodological gap. Global financial engineering literature proves that Regime-Switching Models (RSMs) are highly effective in capturing the non-linear, persistent shifts in financial variables that accompany economic cycles, offering a superior method for forecasting risk (Ang and Timmermann, 2012; Hamilton, 1989). However, this methodology is typically conditioned on endogenous financial variables (e.g., GDP growth or asset volatility) (Koopman and Lucas, 2005).

This dissertation proposes to fill this gap by developing an **Exogenous Two-Regime Markov-Switching Model**. This innovative approach will link the credit migration process of Zimbabwean MFIs directly to a measurable, exogenous climate index (SPEI). By estimating two distinct credit migration matrices (M_{stress} and M_{normal}) and a probabilistic Regime Transition Matrix from climate history, the model transforms a non-financial physical risk into a quantifiable, dynamic financial metric. This framework directly addresses the IFRS 9 requirement by producing a forward-looking, climate-adjusted Expected Credit Loss (ECL) estimate, providing a novel and rigorous solution to a critical problem in emerging market finance.

1.2 Statement of the Problem

The core problem in the Zimbabwean microfinance sector is the fundamental conflict between the dynamic, non-linear reality of Physical Climate Risk (Drought) and the static assessment methodologies used for credit risk management. Global standards, mandated by IFRS 9 and reinforced by regulators like the NGFS, require financial institutions to move beyond backward-looking models to produce forward-looking Expected Credit Loss (ECL) estimates that dynamically incorporate future climate-driven conditions (European Banking Authority, 2023; IASB, 2014). However, Microfinance Institutions (MFIs) in Zimbabwe, whose client base is highly vulnerable to agricultural and climatic shocks, rely on time-homogeneous (TTC) credit risk models. These static models assume average default probabilities, systematically failing to capture the systemic, highly covariant jump in defaults that occurs when the system switches into a severe Drought-Stress Regime.

This methodological failure leads to two critical financial risks:

1. **Chronic underestimation** of actual credit loss exposure during periods of climate stress, and

2. **Inadequate capital provisioning**, which destabilizes the MFI sector and threatens financial inclusion.

While the financial engineering literature offers Regime-Switching Models (RSMs) as the appropriate solution for dynamic volatility (Koopman and Lucas, 2005), the critical and specific technical gap is the absence of an **Exogenous Two-Regime Markov-Switching Model** that translates a measurable, localized climate index (SPEI) into a quantifiable, probabilistic switch in the MFI's credit migration process.

The problem, therefore, is the lack of a mathematically rigorous, dynamic model capable of translating the probability of a physical climate event into an IFRS 9-compliant ECL forecast for the Zimbabwean microfinance sector.

1.3 Research Objectives

The primary objectives of this research are:

1. **Regime Classification:** To establish a statistical methodology to define and historically classify the two credit risk regimes—Normal/Wet and Drought/Stress—using a localized climate index.
2. **Matrix Estimation:** To use Maximum Likelihood Estimation (MLE) to calculate the two distinct Credit Migration Matrices (M_1 and M_2) and the dynamic Regime Transition Matrix.
3. **Model Validation:** To perform a Likelihood Ratio Test (LRT) to statistically validate that the two-regime model provides a significant mathematical improvement over a static, single-matrix model.
4. **Risk Forecasting:** To apply the resulting Drought-Adjusted Regime-Switching Model to dynamically forecast the Probability of Default (PD) and calculate the Expected Credit Loss (ECL).

1.4 Research Questions

This research seeks to answer the following questions:

1. How can a localized climate index (SPEI) be statistically modeled to accurately delineate and classify the historical time series into the distinct Normal-Stress and Drought-Stress credit regimes for Zimbabwean MFIs?
2. What are the quantitative differences between the estimated Normal Regime Migration Matrix and the Drought-Stress Migration Matrix, and how significantly does the Probability of Default (PD) increase between the two

regimes?

3. How can the Exogenous Two-Regime Markov-Switching Model be mathematically calibrated using MLE to estimate the matrix which links the dynamics of the climate index to the probability of switching between the two credit regimes?
4. To what extent does the Drought-Adjusted Regime-Switching Model provide a more accurate and robust estimation of Expected Credit Loss (ECL), compared to traditional time-homogeneous static models, for capital planning and IFRS 9 compliance over a five-year horizon?

1.5 Research Hypothesis

1.5.1 Null Hypothesis (H_0)

The incorporation of the Exogenous Two-Regime Markov-Switching structure does not significantly improve the fit or predictive accuracy of the credit risk model for Zimbabwean MFIs. Therefore, the Probabilities of Default (PD) and resulting Expected Credit Loss (ECL) forecasts are statistically no more accurate than those produced by a traditional static (time-homogeneous) model.

1.5.2 Alternative Hypothesis (H_1)

The Exogenous Two-Regime Markov-Switching Model provides a statistically significant and robust improvement in modeling credit risk by proving that the credit migration parameters are conditional on the external climate regime, resulting in a more accurate and compliant forecast of Expected Credit Loss (ECL) than static models.

1.6 Significance of the Study

The study provides a significant contribution by bridging a critical methodological gap at the intersection of quantitative finance, climate risk, and emerging market regulatory compliance.

1.6.1 Academic Contribution

The central significance is the advancement in stochastic modeling achieved by developing and rigorously validating the Exogenous Two-Regime Markov-Switching Model. This model extends the established frameworks of credit risk modeling

(Koopman and Lucas, 2005; Bangia et al., 2002) by successfully proving and estimating the necessity of an exogenous, non-financial driver (the Drought Index) to model credit volatility. By formally rejecting the simplistic assumption of time-homogeneity ($M_{\text{stress}} = M_{\text{normal}}$) via the Likelihood Ratio Test (LRT), the research provides the academic and statistical proof needed to transition the MFI sector from static TTC (Through-the-Cycle) to dynamic PIT (Point-in-Time) risk assessment.

1.6.2 Practical Contribution

Practically, this research delivers a demonstrably more accurate and robust Expected Credit Loss (ECL) forecast, directly assisting Zimbabwean Microfinance Institutions in achieving mandatory IFRS 9 compliance and enabling the Reserve Bank of Zimbabwe (RBZ) to monitor systemic climate risk using quantifiable, forward-looking parameters.

1.6.3 Policy Implications

Ultimately, the study provides a critical, implementable tool that translates physical climate uncertainty into a manageable financial metric, supporting both capital resilience and sustainable green finance initiatives in the region.

1.7 Scope of the Study

1.7.1 Geographic Scope

This study focuses on Microfinance Institutions (MFIs) operating within Zimbabwe, with particular emphasis on regions with high agricultural dependency and climate sensitivity.

1.7.2 Temporal Scope

The analysis covers historical credit transition data and climate indices spanning a minimum of 10 years to ensure adequate representation of both normal and drought regimes.

1.7.3 Modeling Framework

The study applies the Exogenous Two-Regime Markov-Switching Model, where key parameters—specifically the credit migration matrices—are conditional on an observable drought index (SPEI). This framework captures regime-dependent credit behavior characteristic of Zimbabwe's climate-sensitive economy.

1.7.4 Product Focus

The analysis is restricted to microfinance loan portfolios, particularly those extended to agricultural and informal sector borrowers whose repayment capacity is directly linked to climate conditions.

1.7.5 Comparative Analysis

A rigorous comparison is conducted between:

- The single-regime (time-homogeneous) migration matrix model, and
- The proposed two-regime Markov-Switching model conditioned on drought indices.

1.8 Assumptions of the Study

Before commencing the research, several foundational assumptions guided the study design:

1. **Markov Property:** Credit rating transitions follow a first-order Markov process, where the probability of transitioning to a future state depends only on the current state and the prevailing climate regime.
2. **Exogeneity of Climate Index:** The drought index (γ) is exogenous to the credit system—that is, borrower defaults do not influence the climate, ensuring valid causal inference.
3. **Regime Stationarity:** The two credit migration matrices (M_{normal} and M_{stress}) are stationary within each regime over the sample period.
4. **Data Availability:** Sufficient historical credit transition data and corresponding climate indices are available for robust parameter estimation.
5. **Absorbing Default State:** The default state (Rating 1) is treated as absorbing—once a borrower defaults, they do not recover within the same loan cycle.

1.9 Limitations of the Study

The research acknowledges the following limitations:

1. **Data Sparsity:** Limited historical observations for extreme rating transitions may affect the precision of estimated migration probabilities. Bayesian shrinkage or pooling techniques may be required.
2. **Geographic Generalization:** Results are specific to Zimbabwean MFIs and may not be directly generalizable to other countries without recalibration.
3. **Measurement Error in Climate Index:** Potential misalignment between district-level drought indices and actual borrower locations may introduce measurement error.
4. **Computational Intensity:** Bootstrap inference with large replications requires significant computational resources, potentially limiting the scope of sensitivity analysis.

1.10 Organization of the Study

This dissertation is organized into five chapters:

Chapter 1: Introduction Presents the background, problem statement, research objectives and questions, hypotheses, significance, scope, assumptions, and limitations of the study.

Chapter 2: Literature Review Provides a comprehensive review of theoretical and empirical literature on credit risk modeling, regime-switching models, climate risk integration, and IFRS 9 ECL frameworks.

Chapter 3: Research Methodology Details the data sources, model specification, maximum likelihood estimation procedures, hypothesis testing framework, and ECL calculation methodology.

Chapter 4: Data Analysis and Presentation Presents the empirical results including estimated migration matrices, likelihood ratio tests, SVDM metrics, and ECL forecasts under various scenarios.

Chapter 5: Conclusions and Recommendations Summarizes research findings, draws conclusions, discusses policy implications, and provides recommendations for practitioners and future research.

1.11 Chapter Summary

This chapter introduced the research by establishing the critical need for dynamic, climate-sensitive credit risk models in the Zimbabwean microfinance sector. The global IFRS 9 mandate requires forward-looking ECL estimation, yet local MFIs continue to rely on static models that fail to capture the systemic impact of drought on credit portfolios. The proposed Exogenous Two-Regime Markov-Switching Model addresses this gap by linking credit migration directly to observable drought indices, producing regime-dependent probabilities of default and IFRS 9-compliant ECL forecasts.

The research objectives, questions, and hypotheses have been clearly articulated to guide the empirical investigation. The significance of the study lies in its contribution to both academic methodology and practical risk management for emerging market financial institutions. The subsequent chapter provides a comprehensive review of the theoretical and empirical literature underpinning this research.

CHAPTER 2

Literature Review

2.1 Introduction

This chapter provides a comprehensive review of the theoretical and empirical literature underpinning the development of a drought-adjusted, two-regime Markov-switching credit migration model for Expected Credit Loss (ECL) estimation in Zimbabwean microfinance institutions. The review synthesizes literature from four interconnected domains: (1) credit risk modeling and migration matrices, (2) regime-switching models in finance, (3) climate risk and financial stability, and (4) the IFRS 9 ECL framework.

The chapter begins by presenting the conceptual framework that guides this research, establishing the theoretical relationships between climate variables, credit migration dynamics, and ECL outcomes. Subsequently, it examines the theoretical foundations of credit risk measurement, Markov chain models, and regime-switching methodologies. The empirical literature section critically evaluates prior studies on climate-credit linkages in emerging markets, with particular attention to Sub-Saharan Africa. Finally, the chapter concludes with a justification of the present study, identifying the specific methodological gaps that this research addresses.

2.2 Conceptual Framework

The conceptual framework illustrated in Figure 2 demonstrates how exogenous climate conditions, measured through drought indices, influence the credit migration process of microfinance borrowers, ultimately determining Expected Credit Loss outcomes.

2.2.1 Dependent Variable: Expected Credit Loss (ECL)

The primary output of this study is the Expected Credit Loss (ECL), computed in accordance with IFRS 9 requirements. ECL represents the probability-weighted estimate of credit losses over the expected life of a financial instrument, calculated as the product of Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD), discounted to present value (IASB, 2014). The

ECL metric serves as the foundation for loan loss provisioning, capital adequacy assessment, and regulatory compliance (Novotny-Farkas, 2016).

2.2.2 Independent Variables

2.2.2.1 Drought Index (γ)

The Standardized Precipitation Evapotranspiration Index (SPEI) serves as the primary exogenous climate variable. Introduced by Vicente-Serrano, Beguería and López-Moreno (2010), the SPEI extends the earlier Standardized Precipitation Index (SPI) of McKee, Doesken and Kleist (1993) by incorporating temperature-driven evapotranspiration demand alongside precipitation data. This dual-input formulation makes the SPEI superior for agricultural drought monitoring, as crop water stress depends on the balance between precipitation supply and atmospheric evaporative demand—a distinction that is critical under climate change, where rising temperatures amplify drought severity even without precipitation declines (Vicente-Serrano, Beguería and López-Moreno, 2010). Negative SPEI values indicate drought conditions, with values below -1.0 representing moderate drought and values below -2.0 indicating extreme drought (World Meteorological Organization, 2012). The SPEI has been widely adopted for agricultural drought monitoring in Sub-Saharan Africa due to its improved agricultural alignment and statistical interpretability.

2.2.2.2 Credit Migration Matrix

The credit migration matrix captures the transition probabilities between credit rating states over a defined time horizon. Following the seminal work of Jarrow, Lando and Turnbull (1997), the matrix $P = [p_{ij}]$ represents the probability of transitioning from rating state i to state j over one period. In this framework, the migration matrix is not constant but conditional on the prevailing climate regime, yielding two distinct matrices: M_{normal} for normal/wet conditions and M_{stress} for drought/stress conditions.

2.2.2.3 Regime-Switching Mechanism

The regime weight function $\omega(\gamma_t)$ determines the probability of operating under drought-stress conditions given the observed drought index. This logistic specification, introduced by Hamilton (1989) in the context of business cycle modeling, enables smooth transitions between regimes while maintaining interpretability.

Conceptual Framework: Climate-Conditioned Credit Risk Model

Exogenous Driver Drought Index (γ)	$\longrightarrow$	**Regime Classification** Normal vs. Stress
	$\downarrow$	
Migration Dynamics $P(\gamma) = \omega M_{\text{stress}} + (1-\omega) M_{\text{normal}}$	$\longleftarrow$	**Regime Weight** $\omega(\gamma)$
	$\downarrow$	
Risk Metrics PD, LGD, EAD	$\longrightarrow$	**Regulatory Compliance** ECL (IFRS 9)

Figure 2: Conceptual Framework for Drought-Adjusted ECL Modeling

2.3 Theoretical Literature

2.3.1 Credit Risk Measurement: Foundational Concepts

Credit risk, defined as the potential for financial loss arising from a borrower's failure to meet contractual obligations, represents the most significant risk faced by lending institutions (Bessis, 2015). The measurement and management of credit risk has evolved substantially since the pioneering work of Altman (1968), who introduced the Z-score model for bankruptcy prediction. Contemporary credit risk frameworks distinguish between two fundamental approaches: structural models and reduced-form models.

2.3.1.1 Structural Models

Structural models, originating from the seminal contribution of Merton (1974), conceptualize default as an economic event occurring when a firm's asset value falls below its debt obligations. The Merton model treats equity as a call option on the firm's assets, enabling the derivation of default probabilities from observable market data. Extensions by Black and Cox (1976) introduced first-passage-time models, where default occurs at the first instance when asset value crosses a threshold, rather than solely at maturity. While theoretically elegant, structural models require detailed firm-level balance sheet information, limiting their applicability to microfinance borrowers who typically lack audited financial statements (Crouhy, Galai and Mark, 2000).

2.3.1.2 Reduced-Form Models

Reduced-form or intensity-based models, developed by Jarrow and Turnbull (1995) and Duffie and Singleton (1999), treat default as an exogenous stochastic process governed by a hazard rate or default intensity. Unlike structural models, reduced-form approaches do not require explicit modeling of the borrower's asset dynamics, making them more suitable for contexts where balance sheet data is unavailable. The hazard rate may be specified as a function of observable covariates, including macroeconomic and environmental factors, providing the theoretical foundation for climate-conditioned credit risk modeling (Lando, 2004).

2.3.2 Credit Migration Matrices and Markov Chains

2.3.2.1 The Markov Property in Credit Risk

The application of Markov chain theory to credit risk modeling was formalized by Jarrow, Lando and Turnbull (1997), who demonstrated that credit rating transitions can be modeled as a discrete-time, discrete-state Markov process. Under the Markov assumption, the probability of transitioning to a future rating state depends solely on the current state, not on the historical path of transitions. This memoryless property, while a simplification of reality, enables tractable computation of multi-period default probabilities through matrix multiplication (Israel, Rosenthal and Wei, 2001).

Mathematically, for a K-state rating system with transition matrix $P = [p_{ij}]$, the n-step transition probabilities are given by:

$$P^{(n)} = P^n \tag{1}$$

The cumulative default probability over n periods for a borrower initially in state i is simply the $(i, 1)$ element of P^n, assuming state 1 represents default.

2.3.2.2 Estimation of Migration Matrices

The standard approach to estimating migration matrices involves counting observed transitions over a sample period. Let n_{ij} denote the number of observed transitions from state i to state j. The maximum likelihood estimator (MLE) for the transition probability is:

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k=1}^{K} n_{ik}} \tag{2}$$

This cohort-based approach, employed by major rating agencies, assumes time-homogeneity—that transition probabilities remain constant over the estimation period (Bangia et al., 2002). However, substantial empirical evidence demonstrates that migration probabilities vary systematically with economic conditions, motivating the development of regime-dependent models (Nickell, Perraudin and Varotto, 2000).

2.3.3 Regime-Switching Models in Finance

2.3.3.1 The Hamilton Model

The foundational contribution to regime-switching econometrics is Hamilton's (1989) Markov-switching autoregressive model, developed to capture asymmetric business cycle dynamics in US GDP growth. Hamilton demonstrated that economic time series often exhibit distinct behavioral patterns across expansionary and recessionary regimes, with transitions between regimes governed by an unobserved Markov chain.

The general Markov-switching model specifies:

$$y_t = \mu_{s_t} + \phi(y_{t-1} - \mu_{s_{t-1}}) + \sigma_{s_t}\epsilon_t \tag{3}$$

where $s_t \in \{1, 2, \ldots, R\}$ denotes the unobserved regime at time t, and the regime process follows a first-order Markov chain with transition probabilities $P(s_t = j|s_{t-1} = i) = p_{ij}$.

2.3.3.2 Extensions to Credit Risk

The application of regime-switching frameworks to credit risk gained momentum following the observation that default rates exhibit strong cyclical patterns correlated with macroeconomic conditions. Bangia et al. (2002) estimated separate migration matrices for expansion and contraction regimes, finding substantially higher default probabilities during recessions. Nickell, Perraudin and Varotto (2000) demonstrated that migration matrices vary significantly across industry sectors and business cycle phases, challenging the assumption of unconditional time-homogeneity.

Koopman and Lucas (2005) extended the framework by modeling the regime process as a function of observable macroeconomic indicators, enabling out-of-sample forecasting. Their findings confirmed that conditioning migration matrices on GDP growth and credit spreads substantially improves default prediction accuracy. This approach provides the methodological template for the present study, which substitutes climate indices for macroeconomic indicators as the regime-determining variable.

2.3.3.3 Exogenous vs. Endogenous Regime Determination

A critical distinction in regime-switching models concerns whether regime transitions are determined endogenously (inferred from the data) or exogenously (driven by observable external factors). Standard Hamilton-type models treat regimes as latent states, estimated jointly with other parameters via the Expectation-Maximization algorithm. However, when external regime indicators are available, conditioning on observable covariates offers several advantages: improved interpretability, enhanced forecasting capability, and reduced parameter uncertainty (Krolzig, 1997).

Ang and Timmermann (2012) provide a comprehensive survey of regime-switching methods in finance, emphasizing that exogenous regime indicators—when theoretically justified and empirically validated—yield more stable and interpretable parameter estimates than purely latent specifications. For climate-sensitive credit portfolios, the drought index provides a natural, observable regime indicator with clear economic interpretation.

2.3.4 IFRS 9 Expected Credit Loss Framework

2.3.4.1 Transition from IAS 39 to IFRS 9

The International Financial Reporting Standard 9 (IFRS 9), effective from January 2018, fundamentally reformed the accounting treatment of credit losses. Under the predecessor standard IAS 39, institutions recognized loan losses only after objective evidence of impairment—the so-called "incurred loss" model. This backward-looking approach was criticized for delayed loss recognition during the 2008 Global Financial Crisis, when institutions accumulated substantial unrecognized losses (Novotny-Farkas, 2016).

IFRS 9 replaced the incurred loss model with a forward-looking Expected Credit Loss (ECL) framework. Institutions must now recognize lifetime ECL for assets experiencing "significant increase in credit risk" (SICR) since origination, incorporating "reasonable and supportable information" about future economic conditions (IASB, 2014). This forward-looking mandate explicitly requires models that can translate anticipated macroeconomic and environmental scenarios into credit loss estimates.

2.3.4.2 ECL Calculation Methodology

Under IFRS 9, Expected Credit Loss is computed as the probability-weighted present value of credit losses:

$$\text{ECL} = \sum_{t=1}^{T} \frac{\text{PD}_t \times \text{LGD}_t \times \text{EAD}_t}{(1+r)^t} \tag{4}$$

where PD_t is the marginal probability of default in period t, LGD_t is the loss given default, EAD_t is the exposure at default, and r is the effective interest rate. The IFRS 9 framework distinguishes three stages of credit deterioration, with Stage 1 requiring 12-month ECL and Stages 2-3 requiring lifetime ECL (Cohen and Edwards, 2017).

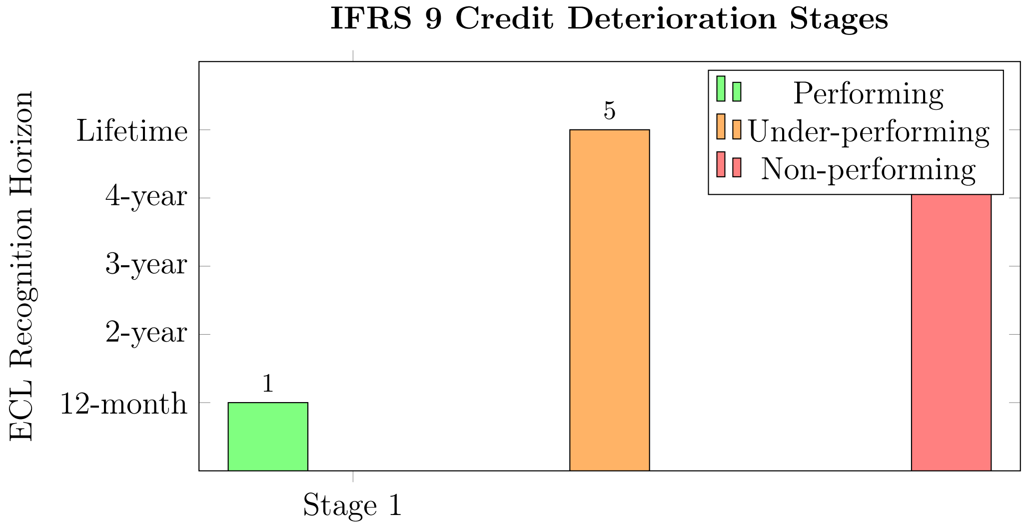

Figure 3: IFRS 9 Three-Stage ECL Model: Stage 1 loans require 12-month ECL provisioning, while Stages 2 and 3 require lifetime ECL recognition

2.3.4.3 Incorporating Forward-Looking Information

A distinguishing feature of IFRS 9 is the requirement to incorporate forward-looking macroeconomic scenarios into ECL estimation. The standard mandates consideration of multiple scenarios, probability-weighted according to their assessed likelihood (IASB, 2014). This requirement has stimulated substantial research on scenario generation and macroeconomic conditioning of credit parameters.

The European Banking Authority (2023) guidance explicitly identifies climate risk as a factor requiring incorporation into ECL models, recognizing that physical climate events can materially impact credit portfolios. This regulatory emphasis provides direct support for the development of drought-adjusted ECL models for climate-vulnerable sectors such as microfinance.

2.4 Empirical Literature

2.4.1 Climate Risk and Credit Performance

2.4.1.1 Global Evidence on Climate-Credit Linkages

The empirical literature on climate-credit linkages has expanded rapidly in response to regulatory pressure and growing recognition of climate change as a systemic financial risk. Faiella and Natoli (2018) analyzed Italian bank loan portfolios and found that firms in flood-prone areas exhibited significantly higher default rates following extreme precipitation events. Noth and Schüwer (2023) examined German banks' exposure to climate-related natural disasters, documenting substantial increases in non-performing loans for banks with concentrated lending in affected regions.

At the macroeconomic level, Klomp (2014) analyzed a panel of 160 countries and found that natural disasters, particularly droughts and floods, significantly increase bank credit risk and reduce financial system stability. The effect is particularly pronounced in developing economies with limited insurance penetration and fiscal capacity. Dafermos, Nikolaidi and Galanis (2018) developed a stock-flow consistent model demonstrating how climate damages propagate through the financial system, amplifying initial shocks through credit contraction and asset price deflation.

2.4.1.2 Evidence from Sub-Saharan Africa

The empirical evidence on climate-credit linkages in Sub-Saharan Africa, while more limited, consistently demonstrates the heightened vulnerability of regional financial institutions to climate shocks. Chikalipah (2018) examined the determinants of loan repayment among Zambian microfinance borrowers, finding that rainfall variability was a significant predictor of default, independent of borrower characteristics and loan terms. The study emphasized the covariant nature of climate risk, which causes simultaneous defaults across the portfolio, undermining the diversification benefits typically assumed in credit risk models.

Collier and Dercon (2014) analyzed the impact of weather shocks on smallholder farmers in Ethiopia and found that drought events led to persistent income losses extending several years beyond the initial shock. These prolonged effects on borrower cash flows imply that drought impacts on credit quality may extend beyond the immediate drought period, supporting the use of regime-switching models that capture persistent state changes.

Castellani and Cincinelli (2015) provided direct evidence on the drought-default relationship in African microfinance, documenting a 40% increase in portfolio-at-risk during severe drought years compared to normal periods. Their analysis of East African MFIs demonstrated that climate-induced credit losses substantially exceeded provisioning levels, threatening institutional solvency.

2.4.2 Credit Risk in Zimbabwean Financial Institutions

2.4.2.1 Macroeconomic Determinants of Credit Risk

The Zimbabwean banking sector has experienced significant volatility, driven by macroeconomic instability, currency crises, and external shocks. Nyamutowa and Masunda (2013) examined the determinants of non-performing loans in Zimbabwean commercial banks, finding that GDP growth, inflation, and lending interest rates were significant macroeconomic predictors of credit quality. However, their study did not incorporate climate variables, despite Zimbabwe's agricultural dependence and vulnerability to drought.

Chigumira and Masiyandima (2021) extended this analysis by examining the relationship between agricultural sector performance and bank credit quality. Their findings confirmed a strong positive correlation between crop production and loan repayment rates, particularly for banks with exposure to agricultural lending. The study recommended that Zimbabwean banks incorporate agricultural output forecasts into their credit risk assessment frameworks.

2.4.2.2 Microfinance Sector Dynamics

The Zimbabwean microfinance sector serves approximately 500,000 active borrowers, predominantly small-scale farmers and informal sector entrepreneurs whose livelihoods are directly dependent on climate conditions (Reserve Bank of Zimbabwe, 2024). Mago and Hofisi (2016) examined the operational challenges facing Zimbabwean MFIs, identifying credit risk management as the most significant constraint to sustainable growth.

The Reserve Bank of Zimbabwe (2024) Microfinance Industry Report documented aggregate portfolio-at-risk (PAR>30 days) of 18.7% for the sector, substantially exceeding the 5% prudential threshold. The report attributed elevated default rates to a combination of macroeconomic challenges and climate shocks, recommending enhanced risk management frameworks that incorporate forward-looking climate information.

2.4.3 Regime-Switching Models in Credit Risk Applications

2.4.3.1 Empirical Performance of Regime-Switching Credit Models

The empirical literature consistently demonstrates that regime-switching credit risk models outperform time-homogeneous alternatives. Bangia et al. (2002) compared migration matrices estimated separately for expansion and contraction periods against an unconditional matrix, finding that the regime-dependent specification substantially improved Value-at-Risk estimates during stress periods.

Koopman, Lucas and Monteiro (2008) developed a regime-switching ordered probit model for credit ratings, with regimes determined by observable macroeconomic indicators. Their analysis of Moody's ratings demonstrated that accounting for regime-dependence reduced root mean squared forecast errors by 15-25% compared to time-homogeneous specifications.

Frydman and Schuermann (2008) examined the stability of credit migration matrices over time, providing statistical tests for detecting regime changes. Their methodology, based on eigenvalue decomposition, has been widely adopted for validating regime-switching specifications in credit risk applications.

2.4.3.2 Applications to Emerging Markets

Regime-switching credit models have been successfully applied to emerging market contexts characterized by high volatility and structural breaks. Leow and Mues (2012) developed a Markov-switching model for UK retail credit, demonstrating improved default prediction during the 2008 financial crisis. Du and Bhattacharyya (2021) applied regime-switching methods to Chinese corporate bonds, finding that conditioning on monetary policy regime substantially improved credit spread forecasting.

However, applications of regime-switching methods to climate-conditioned credit risk in emerging markets remain scarce. Volz et al. (2020) surveyed central banks in 24 developing countries and found that while 85% recognized climate risk as a threat to financial stability, fewer than 20% had implemented quantitative frameworks for assessing climate-related credit risk. This gap motivates the development of practical, implementable models suitable for emerging market contexts.

2.4.4 Drought Indices and Agricultural Credit Risk

2.4.4.1 Drought Index Methodologies

The Standardized Precipitation Index (SPI), developed by McKee, Doesken and Kleist (1993), has become the international standard for drought monitoring due to its computational simplicity, temporal flexibility, and statistical interpretability. The SPI transforms precipitation totals into a standard normal distribution, where negative values indicate below-normal precipitation and positive values indicate above-normal precipitation. SPI values below −1.0, −1.5, and −2.0 correspond to moderate, severe, and extreme drought, respectively (World Meteorological Organization, 2012).

The Standardized Precipitation Evapotranspiration Index (SPEI), introduced by Vicente-Serrano, Beguería and López-Moreno (2010), extends the SPI methodology by incorporating temperature-driven evapotranspiration demand. The SPEI is particularly relevant for climate change contexts where rising temperatures may increase drought severity even without precipitation declines.

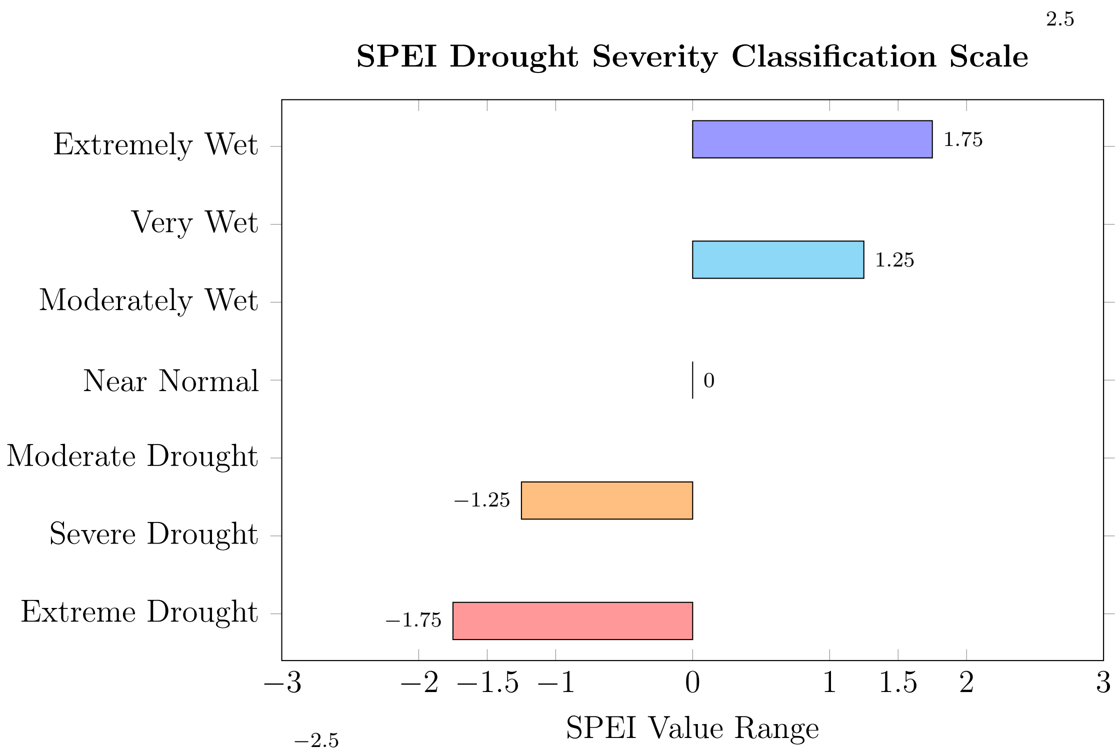

Figure 4: SPEI Drought Severity Classification (World Meteorological Organization, 2012). Negative values indicate drought conditions; values below −1.0, −1.5, and −2.0 correspond to moderate, severe, and extreme drought respectively.

2.4.4.2 Drought Impacts on Agricultural Lending

The empirical relationship between drought indices and agricultural credit performance has been documented across multiple contexts. Turvey and Kong (2010) analyzed Canadian agricultural loans and found that SPI values explained a significant portion of variation in loan default rates, with each standard deviation decrease in SPI associated with a 2.3 percentage point increase in default probability.

Carter et al. (2017) examined the impact of weather index insurance on agricultural lending in Ethiopia and Kenya. Their findings demonstrated that drought shocks led to sharp increases in loan delinquency, but that index insurance provision mitigated these effects by preserving borrower liquidity. The study provides indirect evidence that drought indices effectively capture credit-relevant climate information.

2.5 Justification of the Present Study

The literature review reveals several critical gaps that the present study addresses:

2.5.1 Gap 1: Absence of Exogenous Climate-Conditioned Migration Models

While regime-switching credit risk models are well-established in the literature, existing applications predominantly condition on endogenous financial variables such as GDP growth, credit spreads, or market volatility (Koopman and Lucas, 2005; Bangia et al., 2002). The specific innovation of the present study is the development of a migration model conditioned on an *exogenous* climate variable—the drought index. This approach is particularly appropriate for climate-vulnerable sectors where the causal mechanism runs unambiguously from climate to credit performance, satisfying the exogeneity requirement for valid inference.

2.5.2 Gap 2: Limited Application to African Microfinance

The empirical literature on climate-credit linkages in Sub-Saharan Africa remains nascent, with most studies employing descriptive or reduced-form methodologies (Chikalipah, 2018; Castellani and Cincinelli, 2015). The present study advances this literature by developing a fully specified structural model that generates quantitative estimates of regime-dependent migration matrices and forward-looking ECL forecasts suitable for practical implementation.

2.5.3 Gap 3: IFRS 9 Compliance Gap

Despite the IFRS 9 mandate for forward-looking ECL estimation, practical guidance on incorporating climate scenarios remains limited. The European Banking Authority (2023) has emphasized the need for climate-sensitive credit risk models, but implementation examples for emerging market institutions are scarce. This study provides a rigorous, implementable framework that directly addresses IFRS 9 requirements by translating climate scenarios into ECL estimates.

2.5.4 Gap 4: Zimbabwean Context

No prior study has developed a climate-conditioned credit risk model specifically for Zimbabwean financial institutions, despite the country's well-documented vulnerability to drought and the demonstrated climate sensitivity of its agricultural sector (Chigumira and Masiyandima, 2021). The present study fills this geographic gap by calibrating the model to Zimbabwean MFI data and drought indices, generating context-specific parameter estimates and forecasts.

2.6 Chapter Summary

This chapter has provided a comprehensive review of the theoretical and empirical literature underpinning the development of a drought-adjusted, two-regime Markov-switching credit migration model. The conceptual framework established the theoretical linkages between exogenous climate conditions, regime-dependent migration dynamics, and IFRS 9-compliant ECL estimation.

The theoretical literature review traced the evolution of credit risk measurement from structural and reduced-form models to Markov chain migration matrices and regime-switching specifications. The IFRS 9 framework was examined in detail, emphasizing the regulatory mandate for forward-looking, scenario-conditioned ECL estimation. The empirical literature demonstrated consistent evidence of climate-credit linkages across global and African contexts, while highlighting the absence of fully specified, exogenous regime-switching models for climate-vulnerable microfinance portfolios.

Four critical research gaps were identified: (1) the absence of exogenous climate-conditioned migration models, (2) limited structural modeling for African microfinance, (3) inadequate practical guidance for IFRS 9 climate integration, and (4) the specific Zimbabwean context gap. The present study addresses each of these gaps by developing, estimating, and validating a drought-adjusted Markov-switching ECL framework calibrated to Zimbabwean MFI data.

The subsequent chapter presents the research methodology, detailing the data sources, model specification, estimation procedures, and validation framework employed in this study.

CHAPTER 3

Research Methodology

3.1 Overview of Methodology

This chapter outlines the financial engineering methodology for modeling physical climate risk in microfinance. The core objective is to estimate and forecast Expected Credit Loss (ECL) under IFRS 9 using a two-regime exogenous Markov-switching migration matrix conditioned on a drought index γ (Standardized Precipitation Evapotranspiration Index, SPEI). The methodology links observed quarterly rating transitions to the contemporaneous drought regime, generating two distinct migration matrices: $\mathbf{M}_{\text{normal}}$ (wet/favorable conditions) and $\mathbf{M}_{\text{stress}}$ (drought/stress conditions).

The framework estimates regime-switching probabilities as a function of γ, enabling dynamic ECL forecasting that incorporates both current drought conditions and forward-looking climate scenarios. The methodology implements maximum likelihood estimation, hypothesis testing, and bootstrap inference to quantify parameter uncertainty, ultimately producing probabilistic ECL forecasts suitable for stress testing and provisioning.

3.2 Data, Sample, and Preprocessing

3.2.1 Data Sources

The empirical analysis draws on two primary data streams:

1. **Credit Transition Data:** Quarterly borrower-level credit observations spanning 2001Q1 to 2025Q4 were sourced from regulatory returns filed by licensed Microfinance Institutions with the Reserve Bank of Zimbabwe (RBZ). The dataset covers six provinces—Mashonaland Central, Mashonaland East, Manicaland, Masvingo, Matabeleland South, and Midlands—representing the principal agricultural lending zones. Each observation records the borrower identifier, loan identifier, branch and district, internal credit rating (1–5 scale), outstanding principal balance, and default status.
2. **Climate Data:** The Standardized Precipitation Evapotranspiration Index (SPEI) at quarterly resolution was obtained from the Global SPEI Database

(Vicente-Serrano, Beguería and López-Moreno, 2010), cross-referenced with precipitation and temperature records from the Zimbabwe Meteorological Services Department (MSD). District-level SPEI values were computed for each province to align with the spatial granularity of the credit data.

3.2.2 Input Data Specification

The primary dataset (`ratings_observations.csv`) contains quarterly snapshots of borrower credit positions. The minimal required schema is presented in Table 1.

Table 1: Data Dictionary

Column	**Type**	**Description**	**Example**
borrower_id	string	Unique borrower identifier	MF01234
loan_id	string	Unique loan identifier	MF01234_L001
branch_id	string	Branch code	BR005
district	string	Zimbabwe district or agro-ecological zone	Masvingo
obs_date	date	Quarter-end date (YYYY-MM-DD)	2015-12-31
obs_period	string	Quarter identifier (YYYYQn)	2015Q4
rating	integer	Internal rating (1=worst/default, 5=best)	3
loan_amount	float	Outstanding principal balance (USD or ZWL)	1250.75
default_flag	binary	1 if loan is in default/written-off, 0 otherwise	0
gamma	float	Quarterly drought index (SPEI, mean≈0, $\sigma \approx 1$)	-1.45

3.2.3 Data Processing for Transition Analysis

Let $i_{n,t}$ denote the rating of borrower n at quarter t. For each borrower with consecutive quarterly observations, we extract **one-quarter transitions** ($i_t \to j_{t+1}$) paired with the drought index γ_t at the beginning of the transition period.

Mathematical Notation

- N: Total number of borrowers
- T_n: Number of quarterly observations for borrower n
- Transition set: $\{(i_{n,t}, j_{n,t+1}, \gamma_{n,t})\}$ for $n = 1, \ldots, N$ and $t = 1, \ldots, T_n - 1$
- Exit/repayment: When a borrower repays fully with no default (default_flag = 0, loan_amount = 0), the observation is censored (no transition out of the portfolio).
- Default: Rating 1 is treated as an absorbing state. Once $i_t = 1$, the borrower remains in state 1 unless exited from the portfolio.

Cleaning Rules

1. Align all observations to calendar quarter ends (Mar 31, Jun 30, Sep 30, Dec 31).
2. Handle missing quarters by interpolation only if the gap is exactly one quarter; otherwise, treat as separate spells.
3. Convert all currency amounts to a common real unit (e.g., constant 2025 USD using CPI deflator).
4. Exclude observations with missing critical fields (rating, gamma, loan_amount).

3.3 Model Specification

3.3.1 Conditional Transition Probability

Let $K = 5$ be the number of rating states. The **conditional one-period transition probability** from state i to state j given drought index γ_t is denoted $p_{ij}(\gamma_t)$.

We present two alternative specifications and select the second for primary estimation:

3.3.1.1 Specification 1: Parametric Logistic Link

$$p_{ij}(\gamma_t) = \frac{\exp(\alpha_{ij} + \beta_{ij}\gamma_t)}{\sum_{k=1}^{K} \exp(\alpha_{ik} + \beta_{ik}\gamma_t)} \tag{5}$$

where α_{ij} and β_{ij} are parameters to estimate, with normalization $\alpha_{iK} = \beta_{iK} = 0$ for identifiability (setting state K as reference). This specification allows smooth variation of transition probabilities with γ.

3.3.1.2 Specification 2: Two-Regime Mixture (Primary Model)

$$P(\gamma_t) = \omega(\gamma_t)M_{\text{stress}} + (1 - \omega(\gamma_t))M_{\text{normal}} \tag{6}$$

where:

- $M_{\text{normal}} = [m_{ij}^{\text{norm}}]$ is the $K \times K$ migration matrix under normal conditions
- $M_{\text{stress}} = [m_{ij}^{\text{stress}}]$ is the $K \times K$ migration matrix under drought conditions
- $\omega(\gamma_t)$ is the regime weight function, specified as a logistic function:

$$\omega(\gamma_t) = \frac{1}{1 + \exp(-\kappa(\gamma_t - \gamma_0))} \tag{7}$$

where $\kappa > 0$ controls the sharpness of regime switching and γ_0 is the threshold parameter (typically negative, e.g., -1.0).

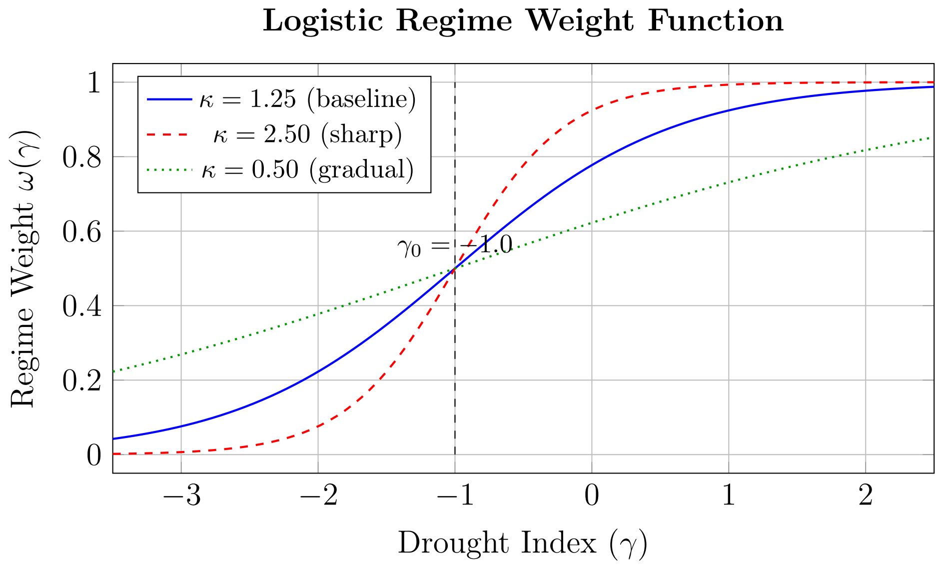

Figure 5: Logistic regime weight function $\omega(\gamma)$ for varying sharpness parameters κ, with threshold $\gamma_0 = -1.0$. As γ decreases (worsening drought), ω increases toward 1, assigning greater weight to M_{stress}.

Constraints: Each row of M_{normal} and M_{stress} must sum to 1, with all entries non-negative.

3.3.2 One-Period Probability of Default (PD)

From any rating state i, the one-period PD is the probability of transitioning to the default state (state 1) in one quarter:

$$\text{PD}_i(\gamma_t) = p_{i1}(\gamma_t) \tag{8}$$

Under the two-regime mixture:

$$\text{PD}_i(\gamma_t) = \omega(\gamma_t) m_{i1}^{\text{stress}} + (1 - \omega(\gamma_t)) m_{i1}^{\text{normal}} \tag{9}$$

3.3.3 Multi-Period Transitions

For forecasting over horizon h quarters, compute the h-step transition matrix via matrix powers:

$$P^{(h)}(\gamma_{t:t+h-1}) = \prod_{s=0}^{h-1} P(\gamma_{t+s}) \tag{10}$$

If regime is assumed constant over horizon (simplifying assumption), then:

$$P^{(h)} = [P(\gamma_t)]^h \tag{11}$$

The cumulative default probability from state i over horizon h is the $(i, 1)$ element of $P^{(h)}$.

3.4 Maximum Likelihood Estimation (MLE)

3.4.1 Log-Likelihood Function

For the two-regime mixture model, the log-likelihood of observing the set of transitions is:

$$\mathcal{L}(\Theta) = \sum_{n=1}^{N} \sum_{t=1}^{T_n-1} \log \left[\omega(\gamma_{n,t}) m^{\text{stress}}_{i_{n,t},j_{n,t+1}} + (1 - \omega(\gamma_{n,t})) m^{\text{normal}}_{i_{n,t},j_{n,t+1}} \right] \tag{12}$$

where $\Theta = \{M_{\text{normal}}, M_{\text{stress}}, \kappa, \gamma_0\}$ is the parameter vector.

3.4.2 Parameterization and Constraints

To enforce row-sum constraints and non-negativity, we use the **softmax reparameterization** for each row i of each matrix.

3.4.2.1 Absorbing Default State Constraint

Since Rating 1 (default) is an absorbing state, the first row of both migration matrices is explicitly constrained and excluded from estimation:

$$m^{\text{norm}}_{1j} = m^{\text{stress}}_{1j} = \begin{cases} 1 & \text{if } j = 1 \\ 0 & \text{if } j \neq 1 \end{cases} \tag{13}$$

This ensures that once a borrower enters default, they remain in the default state with certainty. No free parameters are estimated for Row 1 of either matrix.

3.4.2.2 Softmax Reparameterization

For the remaining non-default rows ($i = 2, \ldots, K$) of M_{normal}, define unconstrained parameters $\theta_{ij}^{\text{norm}} \in \mathbb{R}$ for $j = 1, \ldots, K-1$, with:

$$m_{ij}^{\text{norm}} = \frac{\exp(\theta_{ij}^{\text{norm}})}{1 + \sum_{k=1}^{K-1} \exp(\theta_{ik}^{\text{norm}})} \quad \text{for } j = 1, \ldots, K-1 \tag{14}$$

$$m_{iK}^{\text{norm}} = \frac{1}{1 + \sum_{k=1}^{K-1} \exp(\theta_{ik}^{\text{norm}})} \tag{15}$$

Similarly for M_{stress} with parameters $\theta_{ij}^{\text{stress}}$. The total number of free parameters is:

- $2 \times (K-1) \times (K-1)$ for the non-default rows of both matrices (Row 1 is constrained)
- 2 for (κ, γ_0)

Total: $2(K-1)^2 + 2 = 2 \times 4^2 + 2 = 34$ parameters for $K = 5$.

3.4.2.3 Bayesian Regularization for Data Sparsity

In regimes with limited transition observations—particularly for rare rating categories during drought periods—the standard MLE may produce unstable or degenerate estimates. To address this, we employ a **Dirichlet prior** on each row of the migration matrices, yielding a Maximum A Posteriori (MAP) estimator.

For each non-default row i of each matrix, we specify a symmetric Dirichlet prior:

$$\boldsymbol{m}_i \sim \text{Dir}(\alpha_1, \alpha_2, \ldots, \alpha_K) \tag{16}$$

The hyperparameters are set following a weakly informative specification:

$$\alpha_j = \begin{cases} 1.0 & \text{for the diagonal element } (j = i), \text{ reflecting prior expectation of rating persistence} \\ 0.5 & \text{for off-diagonal transitions } (j \neq i) \end{cases} \tag{17}$$

The resulting MAP objective augments the log-likelihood with the log-prior:

$$\mathcal{L}_{\text{MAP}}(\Theta) = \mathcal{L}(\Theta) + \sum_{r \in \{\text{norm, stress}\}} \sum_{i=2}^{K} \sum_{j=1}^{K} (\alpha_j - 1) \log m_{ij}^r \tag{18}$$

This Dirichlet regularization acts as pseudo-counts, ensuring that transition probability estimates remain numerically stable even when certain transitions are unobserved in a given regime. As the sample size grows, the prior influence diminishes and the MAP estimate converges to the MLE.

3.4.3 Optimization Algorithm

The MLE optimization procedure is summarized as follows. The complete algorithmic specification is provided in Appendix A.

Given the set of observed transitions $\{(i_{n,t},\ j_{n,t+1},\ \gamma_{n,t})\}$, the optimizer seeks:

$$\hat{\Theta} = \arg\max_{\Theta}\ \mathcal{L}(\Theta) \tag{19}$$

where $\Theta = \{\boldsymbol{\theta}^{\text{norm}},\ \boldsymbol{\theta}^{\text{stress}},\ \kappa,\ \gamma_0\}$ and the unconstrained parameters $\boldsymbol{\theta}$ are mapped to valid transition probabilities via the softmax transformation. The L-BFGS-B algorithm is employed for numerical optimization.

Worked Example (Toy Transitions)

Suppose we have 3 transitions:

1. $(3 \to 4, \gamma = -0.5)$
2. $(4 \to 4, \gamma = +0.8)$
3. $(2 \to 1, \gamma = -2.0)$

With initial parameters yielding:

- M_{normal} row 3: PD $= 0.01$, $P(3 \to 4) = 0.15$
- M_{stress} row 3: PD $= 0.05$, $P(3 \to 4) = 0.10$
- $\kappa = 1.0$, $\gamma_0 = -1.0$

For transition 1:

$$\omega(-0.5) = \frac{1}{1 + \exp(-1 \times (-0.5 + 1))} = \frac{1}{1 + \exp(0.5)} = 0.378$$

Probability $= 0.378 \times 0.10 + (1 - 0.378) \times 0.15 = 0.132$

Log-likelihood contribution $= \log(0.132) = -2.025$

Similarly compute for other transitions and sum.

3.5 Inference and Standard Errors

3.5.1 Observed Information Matrix

Let $\hat{\Theta}$ be the MLE. The observed information matrix is:

$$I(\hat{\Theta}) = -\frac{\partial^2 \mathcal{L}}{\partial \Theta \partial \Theta'} \tag{20}$$

evaluated at $\hat{\Theta}$. The asymptotic variance-covariance matrix is:

$$\text{Var}(\hat{\Theta}) \approx I(\hat{\Theta})^{-1} \tag{21}$$

Standard errors are the square roots of the diagonal elements.

3.5.2 Non-Parametric Bootstrap (Recommended)

Due to potential small-sample bias and non-normalities, we implement a **clustered bootstrap** that resamples borrowers with replacement, preserving all time-series observations for each resampled borrower. The complete bootstrap algorithm is detailed in Appendix B.

For $B = 1{,}000$ replications, the bootstrap yields the following statistics for each parameter θ:

$$\widehat{\text{Bias}} = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^{(b)} - \hat{\theta}_{\text{orig}} \tag{22}$$

$$\widehat{\text{SE}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B} \left(\hat{\theta}^{(b)} - \bar{\theta}\right)^2} \tag{23}$$

$$95\% \text{ CI} = \left[\hat{\theta}_{(0.025)},\ \hat{\theta}_{(0.975)}\right] \tag{24}$$

where $\hat{\theta}^{(b)}$ is the estimate from the b-th bootstrap sample and $\hat{\theta}_{(\alpha)}$ denotes the α-quantile of the bootstrap distribution.

Table 2: Estimated Parameters Template

Parameter	Estimate	Std. Err	95% CI Lower	95% CI Upper	p-value
$m_{\text{norm}}(2,1)$	0.025	0.004	0.018	0.032	<0.001
$m_{\text{norm}}(2,3)$	0.150	0.012	0.128	0.172	<0.001
$m_{\text{stress}}(2,1)$	0.085	0.010	0.067	0.103	<0.001
κ	1.25	0.18	0.92	1.58	<0.001
γ_0	-1.10	0.15	-1.38	-0.82	<0.001

3.6 Model Comparison and Hypothesis Testing

3.6.1 Likelihood Ratio Test (LRT)

Test the null hypothesis H_0: Single homogeneous matrix (no regime switching) vs H_1: Two-regime model.

Let:

- ℓ_{single} = maximized log-likelihood under single matrix ($K \times (K-1)$ parameters)
- ℓ_{two} = maximized log-likelihood under two-regime model ($2K(K-1)+2$ parameters)

Test statistic:

$$LR = 2(\ell_{\text{two}} - \ell_{\text{single}}) \sim \chi^2_{\text{df}} \tag{25}$$

Degrees of freedom: $\text{df} = \text{params}_{\text{two}} - \text{params}_{\text{single}} = [2K(K-1)+2] - [K(K-1)] = K(K-1)+2$.

For $K = 5$: $\text{df} = 5 \times 4 + 2 = 22$.

Worked Example

- $\ell_{\text{single}} = -12{,}450.3$
- $\ell_{\text{two}} = -12{,}210.7$
- $LR = 2(-12{,}210.7 + 12{,}450.3) = 479.2$
- Critical value at $\alpha = 0.05$ for $\chi^2(22)$: 33.92

Since $479.2 > 33.92$, reject H_0 in favor of the two-regime model.

3.6.2 Information Criteria

$$\text{AIC} = 2k - 2\ell, \quad \text{where } k = \text{number of parameters} \tag{26}$$
$$\text{BIC} = k\log(n) - 2\ell, \quad \text{where } n = \text{number of transitions} \tag{27}$$

Lower values indicate better fit with parsimony penalty.

3.6.3 Out-of-Sample Validation

Use rolling window: Estimate on periods 1 to $T-4$, forecast one-quarter transitions for $T-3$ to T, compute out-of-sample log-likelihood. Compare models.

3.7 Measuring Migration Differences (SVDM & Mobility Metrics)

3.7.1 Singular Value Decomposition Metric (SVDM)

To quantify the difference between M_{normal} and M_{stress}, compute:

$$\text{SVDM}(M_1, M_2) = \sum_{k=1}^{K} |\sigma_k(M_1) - \sigma_k(M_2)| \tag{28}$$

where $\sigma_k(\cdot)$ are the singular values in descending order. Larger SVDM indicates greater regime differentiation. The computational procedure for the SVDM and its bootstrap confidence interval is provided in Appendix C.

3.7.2 Bootstrap Confidence Interval for SVDM

Using the B bootstrap replications from Appendix B, the SVDM is recomputed for each pair of bootstrapped matrices $(\hat{M}^{(b)}_{\text{normal}}, \hat{M}^{(b)}_{\text{stress}})$, yielding the empirical distribution $\{\text{SVDM}^{(b)}\}_{b=1}^{B}$. The 95% confidence interval is obtained as:

$$\text{CI}^{\text{SVDM}}_{95\%} = \left[\text{SVDM}_{(0.025)},\ \text{SVDM}_{(0.975)}\right] \tag{29}$$

Table 3: Migration Comparison Template

Metric	$\mathbf{M}_{\text{normal}}$ (avg)	$\mathbf{M}_{\text{stress}}$ (avg)	SVDM	95% CI
PD(2)	0.025	0.085	0.421	[0.312, 0.530]
PD(3)	0.012	0.045	–	–
PD(4)	0.005	0.022	–	–
PD(5)	0.002	0.010	–	–

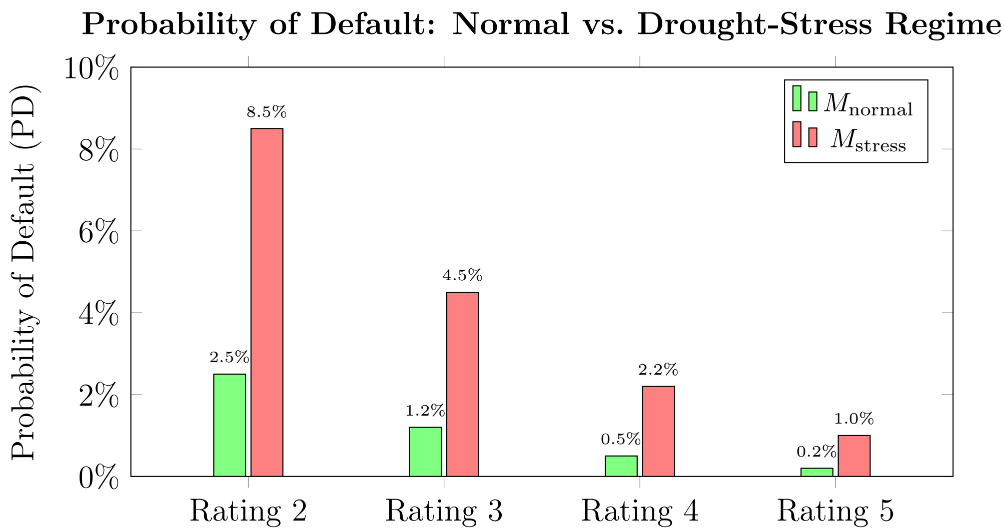

Figure 6: One-quarter Probability of Default by credit rating under the Normal and Drought-Stress regimes. Default probabilities increase by a factor of 3–5× during drought conditions.

3.8 Expected Credit Loss (ECL) Calculation (IFRS 9 Alignment)

3.8.1 ECL Formula

For a loan with current rating i at time t:

One-period ECL

$$\text{ECL}_{t+1} = \text{EAD}_{t+1} \times \text{LGD} \times \text{PD}_i(\gamma_t) \tag{30}$$

where:

- EAD_{t+1} = Exposure at Default (projected outstanding balance at $t+1$)
- LGD = Loss Given Default (assumed 45% unless loan-specific)
- $\text{PD}_i(\gamma_t)$ = one-period probability of default from state i given γ_t

Lifetime ECL (H quarters)

$$\text{Lifetime ECL}_t = \sum_{h=1}^{H} \text{DF}(h) \times \text{EAD}_{t+h} \times \text{LGD} \times \text{PD}_{t\to t+h}(i) \tag{31}$$

where:

- $\text{DF}(h) = (1+r)^{-h/4}$ is quarterly discount factor (annual rate r)
- $\text{PD}_{t\to t+h}(i)$ = cumulative probability of default by horizon h starting from state i:

$$\text{PD}_{t\to t+h}(i) = [P(\gamma_t)^h]_{i,1} \tag{32}$$

under constant regime assumption, or more generally using forward γ paths.

3.8.2 Worked Example

Portfolio:

1. Loan A: Rating 4, EAD = \$1,000, $\gamma_t = -0.3$ (normal regime)
2. Loan B: Rating 3, EAD = \$2,000, $\gamma_t = -1.5$ (stress regime)
3. Loan C: Rating 2, EAD = \$1,500, $\gamma_t = -2.0$ (stress regime)

Assumptions: LGD = 45%, annual discount rate $r = 5\%$, horizon $H = 4$ quarters.

Toy Matrices:

- M_{normal} (row 4): PD = 0.005; M_{stress} (row 4): PD = 0.025
- M_{normal} (row 3): PD = 0.012; M_{stress} (row 3): PD = 0.045
- M_{normal} (row 2): PD = 0.025; M_{stress} (row 2): PD = 0.085

Regime weight: $\omega(-0.3) = 0.43$, $\omega(-1.5) = 0.82$, $\omega(-2.0) = 0.92$

Loan A PD calculation:

$$\text{PD} = 0.43 \times 0.025 + (1 - 0.43) \times 0.005 = 0.0136$$

Cumulative PDs over 4 quarters (assuming regime persists):

Using matrix power P^4 where $P = \omega M_{\text{stress}} + (1-\omega) M_{\text{normal}}$:

- Loan A: $\text{PD}_{\text{cumul}} = 0.053$
- Loan B: $\text{PD}_{\text{cumul}} = 0.165$
- Loan C: $\text{PD}_{\text{cumul}} = 0.285$

Lifetime ECL Calculation (simplified, assuming constant EAD):

$$\mathrm{ECL}_A = 1000 \times 0.45 \times \sum_{h=1}^{4} (1.05)^{-h/4} \times \mathrm{PD}_h$$

where PD_h is marginal default probability in quarter h: $\mathrm{PD}_h = \mathrm{PD}_{\mathrm{cumul}}(h) - \mathrm{PD}_{\mathrm{cumul}}(h-1)$.

Table 4: ECL Forecast Template

Loan	Rating	γ_t	Regime Wt.	1Q PD	4Q Cumul PD	EAD	LGD	Disc. E
A	4	-0.3	0.43	0.0136	0.053	1000	0.45	22
B	3	-1.5	0.82	0.0391	0.165	2000	0.45	136
C	2	-2.0	0.92	0.0802	0.285	1500	0.45	171
Total								**330**

3.9 Forecasting Regimes and Scenario Analysis

3.9.1 Scenario Generation Methods

1. Deterministic γ Paths

- **Baseline:** γ follows seasonal average (historical mean by quarter)
- **Drought scenario:** $\gamma = -1.5$ for next 4 quarters
- **Severe drought:** $\gamma = -2.0$ for next 4 quarters

2. Probabilistic Simulation

The probabilistic simulation generates S climate scenarios by resampling from the historical drought index sequence. For each scenario $s = 1, \ldots, S$, a drought path $\{\gamma_{t+1}^{(s)}, \ldots, \gamma_{t+H}^{(s)}\}$ is drawn via block bootstrap from the observed series, and the corresponding ECL is computed. The full algorithm is provided in Appendix D. The resulting ECL distribution yields:

$$\overline{\mathrm{ECL}} = \frac{1}{S} \sum_{s=1}^{S} \mathrm{ECL}^{(s)} \tag{33}$$

$$\mathrm{VaR}_{95\%} = \mathrm{ECL}_{(0.95)} \tag{34}$$

where $\mathrm{ECL}_{(\alpha)}$ denotes the α-quantile of the simulated ECL distribution.

3. Stochastic Model for γ (Optional AR(1))

$$\gamma_{t+1} = \phi\gamma_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2) \tag{35}$$

Estimated from historical γ series.

3.9.2 Scenario ECL Comparison

Table 5: Scenario ECL Comparison Template

Scenario	**Description**	**Probability**	**1-Year ECL**	**5-Year ECL**
Baseline	Historical average γ	60%	\$1.2M	\$3.8M
Moderate Drought	$\gamma = -1.0$ for 3Q	25%	\$1.8M	\$5.4M
Severe Drought	$\gamma = -2.0$ for 4Q	10%	\$2.7M	\$8.1M
Extreme Wet	$\gamma = +1.5$ for 2Q	5%	\$0.9M	\$2.9M
Expected	Weighted average	**100%**	**\$1.45M**	**\$4.52M**

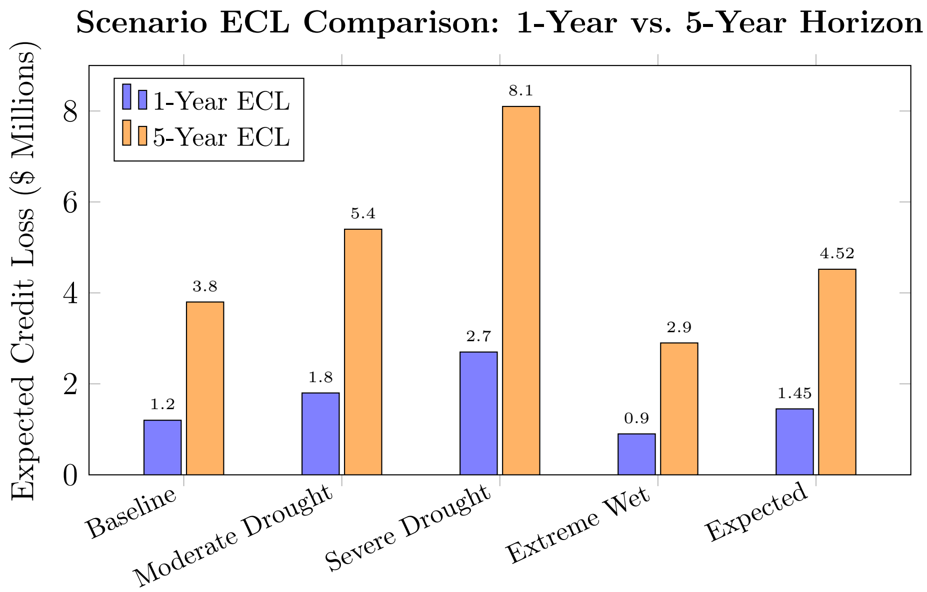

Figure 7: Expected Credit Loss under four climate scenarios and the probability-weighted expected value. Severe drought increases 5-year ECL by 113% relative to the baseline.

3.10 Model Validation, Backtesting and Robustness Checks

3.10.1 Out-of-Sample Backtesting Procedure

The out-of-sample backtesting follows an expanding-window scheme. The data is partitioned into an estimation window (first 80%) and a test window (last 20%). For each quarter t in the test window, the model is re-estimated on all observations up to $t-1$, and one-quarter-ahead PDs are forecast for every active loan at time t. These forecasts are compared against realized default outcomes at $t+1$ to compute discrimination and calibration metrics. The complete backtesting algorithm is provided in Appendix E.

3.10.2 Performance Metrics

1. **Area Under ROC Curve (AUC):** Measures discrimination ability
2. **Brier Score:** $BS = \frac{1}{N}\sum_{n=1}^{N}(p_n - y_n)^2$, where y_n is 0/1 default indicator
3. **Calibration:** Group forecasts into deciles, compare mean predicted PD vs actual default rate

3.10.3 Robustness Checks

1. **LGD Sensitivity:** Recompute ECL with LGD $= 40\%, 45\%, 50\%$
2. **Gamma Lag Specification:** Use γ_{t-1} or $\max(\gamma_t, \gamma_{t-1}, \gamma_{t-2})$
3. **Regional Heterogeneity:** Estimate separate models for drought-sensitive vs resilient regions
4. **Exposure Weighting:** Weight transitions by loan amount in MLE

Table 6: Robustness Analysis Template

Specification	AUC	Brier Score	1-Year ECL ($M)	% Change
Baseline (γ_t)	0.78	0.021	1.45	–
γ_{t-1}	0.76	0.022	1.42	-2.1%
LGD $= 40\%$	0.78	0.021	1.29	-11.0%
Regional Split	0.80	0.020	1.48	$+2.1\%$

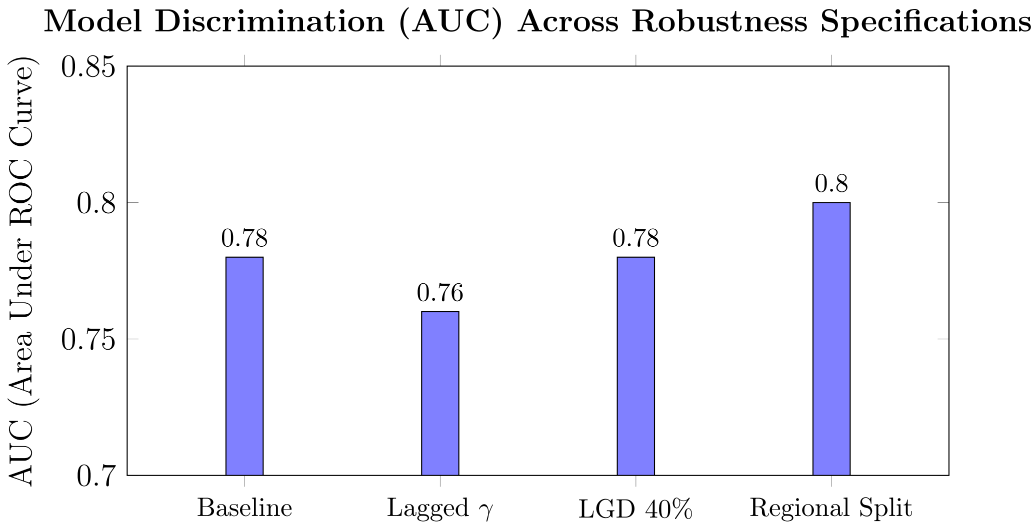

Figure 8: Out-of-sample AUC across robustness specifications. The regional split specification achieves the highest discrimination, while all specifications exceed the 0.70 adequacy threshold.

3.11 Implementation Notes, Assumptions, and Limitations

3.11.1 Key Assumptions

1. **Markov Property:** Rating transitions depend only on current state and γ_t, not on history.
2. **Exogeneity of γ:** Drought index is exogenous to borrower performance (no feedback from defaults to climate).
3. **Stationarity Conditional on Regime:** Migration matrices are constant within regimes over the sample period.
4. **LGD Homogeneity:** LGD is constant across loans and time; sensitivity analysis addresses this.
5. **Absorbing Default:** State 1 (default) is absorbing; no recovery to performing status.

3.11.2 Limitations and Mitigations

1. **Data Sparsity:** Few transitions to/from extreme ratings → Stabilized via Bayesian Dirichlet priors on transition rows (see Section 3.4), which add pseudo-counts to prevent degenerate MLE estimates in low-volume regimes.
2. **Measurement Error in γ:** Potential misalignment between district-level γ and borrower location → Use borrower-weighted average of nearby weather

stations.

3. **Endogeneity Concerns:** Drought may correlate with macroeconomic shocks → Include time fixed effects or macroeconomic controls in extended model.
4. **Non-Stationarity:** Climate change may alter γ distribution → Use rolling windows or time-varying parameters.

3.12 Deliverables & Reproducibility Checklist

3.12.1 Required Outputs

1. Data Files

- `ratings_observations.csv` (cleaned input)
- `transitions_processed.csv` (i, j, gamma, weight for each transition)

2. Parameter Estimates

- `estimated_parameters.csv` (Table 2 format)
- `migration_matrices.csv` (M_{normal} and M_{stress} in readable format)

3. Test Results

- `likelihood_ratio_test.csv` (LR statistic, df, p-value)
- `model_comparison.csv` (AIC, BIC, out-of-sample metrics)
- `svdm_results.csv` (SVDM and bootstrap CI)

4. ECL Forecasts

- `ecl_forecasts_scenarios.csv` (Table 5 format)
- `ecl_time_series.csv` (historical and forecast ECL by quarter)

5. Validation Outputs

- `backtesting_metrics.csv` (AUC, Brier score by quarter)
- `calibration_plot_data.csv` (predicted vs actual default rates by decile)

3.12.2 Folder Structure

The recommended project folder structure for reproducibility is provided in Appendix E.

3.12.3 Implementation Checklist

- ☐ Clean and align quarterly data
- ☐ Extract one-quarter transitions
- ☐ Implement MLE with softmax parameterization
- ☐ Run bootstrap ($B = 1000$) for standard errors
- ☐ Compute Likelihood Ratio Test against single-regime model
- ☐ Calculate SVDM distance between regimes
- ☐ Implement ECL calculator with discounting
- ☐ Generate scenario forecasts (baseline, drought, wet)
- ☐ Perform out-of-sample backtesting
- ☐ Run robustness checks
- ☐ Produce all required output files

3.13 Summary

This chapter has detailed a comprehensive financial engineering methodology for estimating drought-sensitive credit migration matrices and forecasting IFRS 9 Expected Credit Loss. The two-regime Markov-switching framework, conditioned on an exogenous drought index γ, provides a parsimonious yet flexible approach to modeling climate risk in microfinance portfolios. Through maximum likelihood estimation, bootstrap inference, and rigorous validation procedures, the methodology produces statistically robust estimates of regime-dependent migration probabilities.

The ECL forecasting engine incorporates forward-looking climate scenarios, enabling both point estimates and distributional analysis of provisioning requirements. The subsequent chapter will apply this methodology to the Zimbabwean microfinance dataset, presenting empirical estimates, model diagnostics, and ECL forecasts under various drought scenarios.

APPENDICES

A MLE Optimization Algorithm

The maximum likelihood estimation proceeds as follows. Let the observed transition dataset be $\mathcal{D} = \{(i_{n,t},\, j_{n,t+1},\, \gamma_{n,t})\}$ for $n = 1, \ldots, N$ borrowers and $t = 1, \ldots, T_n - 1$ quarters.

Step 1: Parameter Initialization. Define unconstrained parameter matrices $\boldsymbol{\theta}^{\text{norm}} \in \mathbb{R}^{K\times(K-1)}$ and $\boldsymbol{\theta}^{\text{stress}} \in \mathbb{R}^{K\times(K-1)}$, initialized at zero (or small random values). Set initial regime-switching parameters $\kappa^{(0)} = 1.0$ and $\gamma_0^{(0)} = -1.0$.

Step 2: Softmax Transformation. For each row i of each matrix, compute the valid transition probabilities:

$$m_{ij}^{\text{norm}} = \frac{\exp(\theta_{ij}^{\text{norm}})}{1 + \sum_{k=1}^{K-1} \exp(\theta_{ik}^{\text{norm}})}, \quad j = 1, \ldots, K-1 \tag{36}$$

$$m_{iK}^{\text{norm}} = \frac{1}{1 + \sum_{k=1}^{K-1} \exp(\theta_{ik}^{\text{norm}})} \tag{37}$$

and analogously for M_{stress}.

Step 3: Log-Likelihood Evaluation. For each transition (i, j, γ) in $\mathcal{D}$, compute:

$$\omega = \frac{1}{1 + \exp\big(-\kappa(\gamma - \gamma_0)\big)} \tag{38}$$

$$p_{ij}(\gamma) = \omega \cdot m_{ij}^{\text{stress}} + (1-\omega) \cdot m_{ij}^{\text{normal}} \tag{39}$$

The total log-likelihood is accumulated as:

$$\mathcal{L}(\Theta) = \sum_{(i,j,\gamma)\in\mathcal{D}} \log\big[p_{ij}(\gamma)\big] \tag{40}$$

Step 4: Numerical Optimization. Maximize $\mathcal{L}(\Theta)$ with respect to $\Theta = \{\boldsymbol{\theta}^{\text{norm}}, \boldsymbol{\theta}^{\text{stress}}, \kappa, \gamma_0\}$ using the L-BFGS-B algorithm (a quasi-Newton method for bound-constrained optimization). Convergence is assessed by monitoring the gradient norm $\|\nabla\mathcal{L}\| < \varepsilon$ with tolerance $\varepsilon = 10^{-6}$.

Step 5: Output. The estimated parameters $\hat{\Theta}$ yield the migration matrices $\hat{M}_{\text{normal}}$, $\hat{M}_{\text{stress}}$, regime-switching parameters $(\hat{\kappa}, \hat{\gamma}_0)$, and the maximized log-likelihood $\mathcal{L}(\hat{\Theta})$.

B Non-Parametric Bootstrap Inference Procedure

The clustered bootstrap procedure for computing standard errors and confidence intervals proceeds as follows.

Step 1: Original Estimation. Estimate the full model on the complete dataset $\mathcal{D}$ to obtain the point estimates $\hat{\Theta}_{\text{orig}}$.

Step 2: Bootstrap Resampling. For each replication $b = 1, \ldots, B$ (where $B = 1{,}000$):

1. Draw a random sample of N borrower identifiers *with replacement* from the set of all borrower identifiers $\{1, \ldots, N\}$.
2. Construct the bootstrap dataset $\mathcal{D}^{(b)}$ containing *all* quarterly transition records associated with the resampled borrowers.
3. Re-estimate the model on $\mathcal{D}^{(b)}$ using the MLE procedure (Appendix A) to obtain $\hat{\Theta}^{(b)}$.

Step 3: Bootstrap Statistics. For each scalar parameter $\theta \in \Theta$, compute:

$$\widehat{\text{Bias}}(\theta) = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^{(b)} - \hat{\theta}_{\text{orig}} \tag{41}$$

$$\widehat{\text{SE}}(\theta) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B} \left(\hat{\theta}^{(b)} - \bar{\theta}\right)^2} \tag{42}$$

$$95\%\ \text{CI}(\theta) = \left[\hat{\theta}_{(0.025)},\ \hat{\theta}_{(0.975)}\right] \tag{43}$$

where $\bar{\theta} = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^{(b)}$ is the bootstrap mean and $\hat{\theta}_{(\alpha)}$ denotes the α-quantile of the empirical bootstrap distribution.

C SVDM Calculation and Bootstrap Confidence Interval

C.1 SVDM Computation

Given two $K \times K$ migration matrices M_1 and M_2:

1. Compute the singular value decomposition of each matrix: $M_1 = U_1 \Sigma_1 V_1^\top$ and $M_2 = U_2 \Sigma_2 V_2^\top$.
2. Extract the diagonal singular values $\sigma_1(M_1) \geq \sigma_2(M_1) \geq \cdots \geq \sigma_K(M_1)$ and $\sigma_1(M_2) \geq \cdots \geq \sigma_K(M_2)$.

3. Compute the SVDM:

$$\mathrm{SVDM}(M_1, M_2) = \sum_{k=1}^{K} |\sigma_k(M_1) - \sigma_k(M_2)| \tag{44}$$

C.2 Bootstrap Confidence Interval for SVDM

Using the B bootstrap replications from Appendix B:

1. For each replication $b = 1, \ldots, B$, compute $\mathrm{SVDM}^{(b)} = \mathrm{SVDM}\big(\hat{M}^{(b)}_{\text{normal}},\ \hat{M}^{(b)}_{\text{stress}}\big)$.
2. The 95% bootstrap confidence interval is:

$$\mathrm{CI}^{\mathrm{SVDM}}_{95\%} = \big[\mathrm{SVDM}_{(0.025)},\ \mathrm{SVDM}_{(0.975)}\big] \tag{45}$$

where $\mathrm{SVDM}_{(\alpha)}$ is the α-quantile of the empirical bootstrap distribution $\{\mathrm{SVDM}^{(b)}\}_{b=1}^{B}$.

D Probabilistic Scenario Simulation Algorithm

The Monte Carlo simulation for generating the ECL distribution under probabilistic climate scenarios proceeds as follows.

Step 1: Scenario Generation. For each simulation run $s = 1, \ldots, S$:

1. Draw a random starting index $t_0^{(s)} \sim \mathrm{Uniform}\{1, T - H\}$ from the historical drought index series.
2. Extract the block of H consecutive historical observations as the drought path: $\boldsymbol{\gamma}^{(s)} = \{\gamma_{t_0^{(s)}},\ \gamma_{t_0^{(s)}+1},\ \ldots,\ \gamma_{t_0^{(s)}+H-1}\}$.

Step 2: ECL Computation Along Path. For each scenario s, compute the regime-weighted transition matrix at each horizon step $h = 1, \ldots, H$:

$$P_h^{(s)} = \omega\big(\gamma_{t_0^{(s)}+h-1}\big)\, M_{\text{stress}} + \big(1 - \omega\big(\gamma_{t_0^{(s)}+h-1}\big)\big)\, M_{\text{normal}} \tag{46}$$

and accumulate the lifetime ECL:

$$\mathrm{ECL}^{(s)} = \sum_{h=1}^{H} \mathrm{DF}(h) \times \mathrm{EAD}_{t+h} \times \mathrm{LGD} \times \left[\prod_{\ell=1}^{h} P_\ell^{(s)}\right]_{i,1} \tag{47}$$

Step 3: Summary Statistics. From the empirical ECL distribution $\{\mathrm{ECL}^{(s)}\}_{s=1}^{S}$, compute:

$$\overline{\mathrm{ECL}} = \frac{1}{S}\sum_{s=1}^{S} \mathrm{ECL}^{(s)} \tag{48}$$

$$\mathrm{VaR}_{95\%} = \mathrm{ECL}_{(0.95)} \tag{49}$$

$$\mathrm{ES}_{95\%} = \frac{1}{\lfloor 0.05S \rfloor} \sum_{s:\,\mathrm{ECL}^{(s)} \geq \mathrm{VaR}_{95\%}} \mathrm{ECL}^{(s)} \tag{50}$$

E Out-of-Sample Backtesting Algorithm

The expanding-window backtesting procedure is specified as follows.

Step 1: Data Partition. Split the full dataset $\mathcal{D}$ into an initial estimation window $\mathcal{D}_{\text{est}} = \{\text{observations for quarters } 1, \ldots, T_{\text{split}}\}$ (first 80%) and a test window $\mathcal{D}_{\text{test}} = \{\text{observations for quarters } T_{\text{split}} + 1, \ldots, T\}$ (last 20%).

Step 2: Rolling Forecast. For each quarter $t \in \{T_{\text{split}} + 1, \ldots, T - 1\}$:

1. *Estimation:* Re-estimate the model parameters $\hat{\Theta}_t$ using all data from quarters 1 through $t - 1$ via the MLE algorithm (Appendix A).
2. *Forecasting:* For each active loan n with current rating $i_{n,t}$ and observed drought index γ_t, compute the one-quarter-ahead probability of default:

$$\widehat{\mathrm{PD}}_{n,t} = \hat{\omega}(\gamma_t)\, \hat{m}^{\text{stress}}_{i_{n,t},1} + \big(1 - \hat{\omega}(\gamma_t)\big)\, \hat{m}^{\text{normal}}_{i_{n,t},1} \tag{51}$$

3. *Realization:* Observe the actual default indicator $y_{n,t+1} \in \{0, 1\}$ at quarter $t + 1$.

Step 3: Performance Evaluation. Aggregate forecast–realization pairs $\{(\widehat{\mathrm{PD}}_{n,t},\ y_{n,t+1})\}$ across all test quarters and compute the AUC, Brier Score, and calibration metrics as specified in Section 3.10.

Appendix F: Recommended Folder Structure

The following directory layout is recommended to ensure full reproducibility of the estimation, forecasting, and validation pipeline described in Chapter 3.

```
/research/
+-- data/
|   +-- raw/                    % Raw regulatory returns from RBZ
|   +-- processed/              % Cleaned panel dataset (jay_dataset.csv)
|   +-- external/               % SPEI series from Global SPEI Database
+-- estimation/
|   +-- mle_markov.m            % MLE optimisation (Appendix A)
|   +-- bootstrap_inference.m   % Bootstrap standard errors (Appendix B)
|   +-- bayesian_prior.m        % Dirichlet MAP regularisation
+-- testing/
|   +-- svdm_calculator.m       % Stress-normal divergence (Appendix C)
|   +-- robustness_tests.m      % Robustness & sensitivity
+-- forecasting/
|   +-- prob_simulation.m       % Monte-Carlo PD simulation (Appendix D)
|   +-- ecl_scenarios.m         % ECL under macro scenarios
+-- validation/
|   +-- backtesting.m           % Out-of-sample backtesting (Appendix E)
|   +-- auc_brier.m             % Discrimination & calibration metrics
|   +-- chi_square_test.m       % Goodness-of-fit
+-- figures/
    +-- regime_weight.pdf        % Logistic weight function plot
    +-- pd_comparison.pdf        % Normal vs stress PD comparison
    +-- ecl_scenarios.pdf        % Scenario ECL bar chart
```

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